

Streamly — High-Level Design

1. Functional Requirements

- Sign up and log in via email/password or third-party OAuth (Google, Apple)
- Browse titles by category, genre, language, and curated collections
- Search catalog by title keyword, cast member, or genre tag
- View a title's detail page (synopsis, cast, runtime, content rating)
- Stream video with adaptive bitrate based on network conditions
- Resume playback from the last saved position across devices
- Add or remove titles from a personal watchlist
- Support several viewer profiles under a single household account
- Track watch history separately for each profile
- Generate personalized title recommendations per profile
- Allow titles to be downloaded for offline mobile viewing

2. Non-Functional Requirements

Quality	Description / Target
Availability	Platform reachable 99.99% of the time (~52 min downtime/year).
Scalability	Scales horizontally to tens of millions of concurrent viewers at peak.
Low Latency	Playback starts within 2 seconds; API response < 200 ms.
Reliability	No loss of resume position or profile data during failures.
Security	TLS in transit, encryption at rest, bcrypt-hashed passwords, short-lived OAuth tokens.
Durability	Video assets triple-replicated across at least two regions.
Consistency	Strong consistency for billing; eventual consistency acceptable for recommendations.

3. Back-of-the-Envelope Estimation

Metric	Estimate
Total registered users	300 million
Daily Active Users (DAU)	105 million (~35%)
Peak concurrent streams	15 million (~14% of DAU at peak hour)
Average watch time per user/day	~2.1 hours
Average API requests per user/day	~75 (browse, search, sync, downloads)
Peak QPS (API)	~93,000 requests/sec
Average video size (per title, all bitrates)	~7 GB

Metric	Estimate
Total content catalog size	22,000 titles
Peak streaming bandwidth	15M users x 4.5 Mbps avg ≈ 67.5 TB/s (served via CDN)
Total video storage (with replication)	22,000 x 7GB ≈ 154 TB raw; ~462 TB with 3x replication
New content uploaded per day	~90 titles
CDN cache hit ratio (target)	~95%

Assumptions vs. a typical baseline: a larger long-tail catalog (22K titles), a higher average watch time (2.1 hrs), and a slightly lower CDN cache hit ratio because more of the catalog is infrequently accessed.

4. Database Design

USER	PROFILE	WATCHLIST	WATCH_HISTORY	CONTENT
user_id (PK)	profile_id (PK)	profile_id (FK, PK)	history_id (PK)	content_id (PK)
email	user_id (FK)	content_id (FK, PK)	profile_id (FK)	title
password_hash	display_name	added_at	content_id (FK)	genre
created_at	avatar_url		progress_seconds	release_year
	Relationships			

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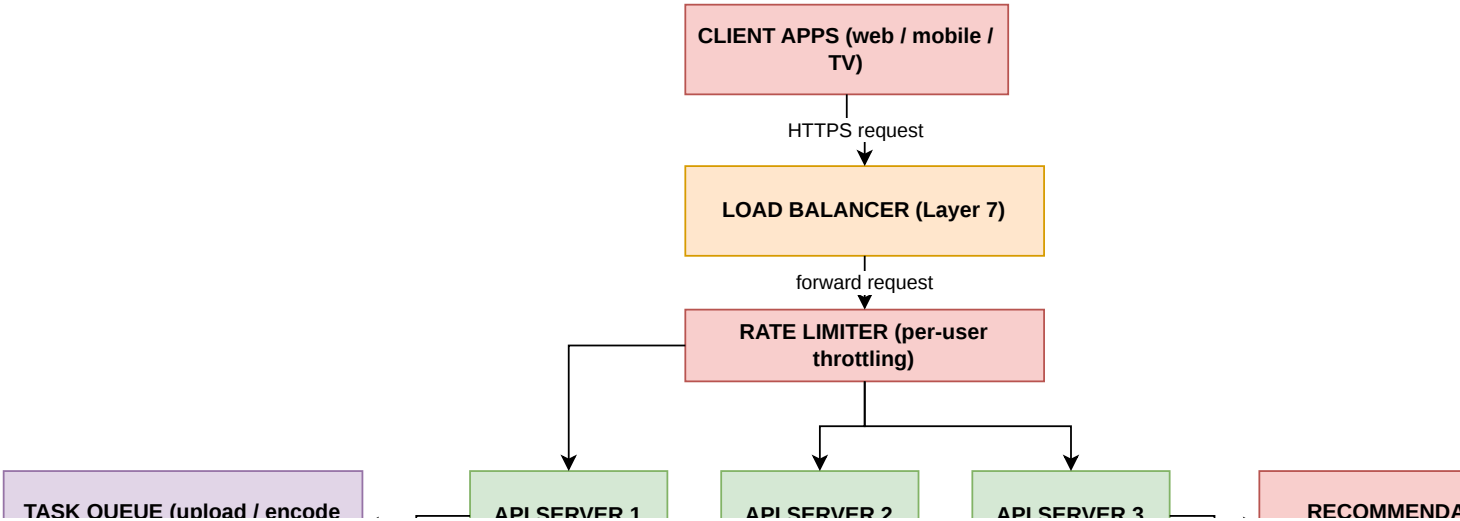
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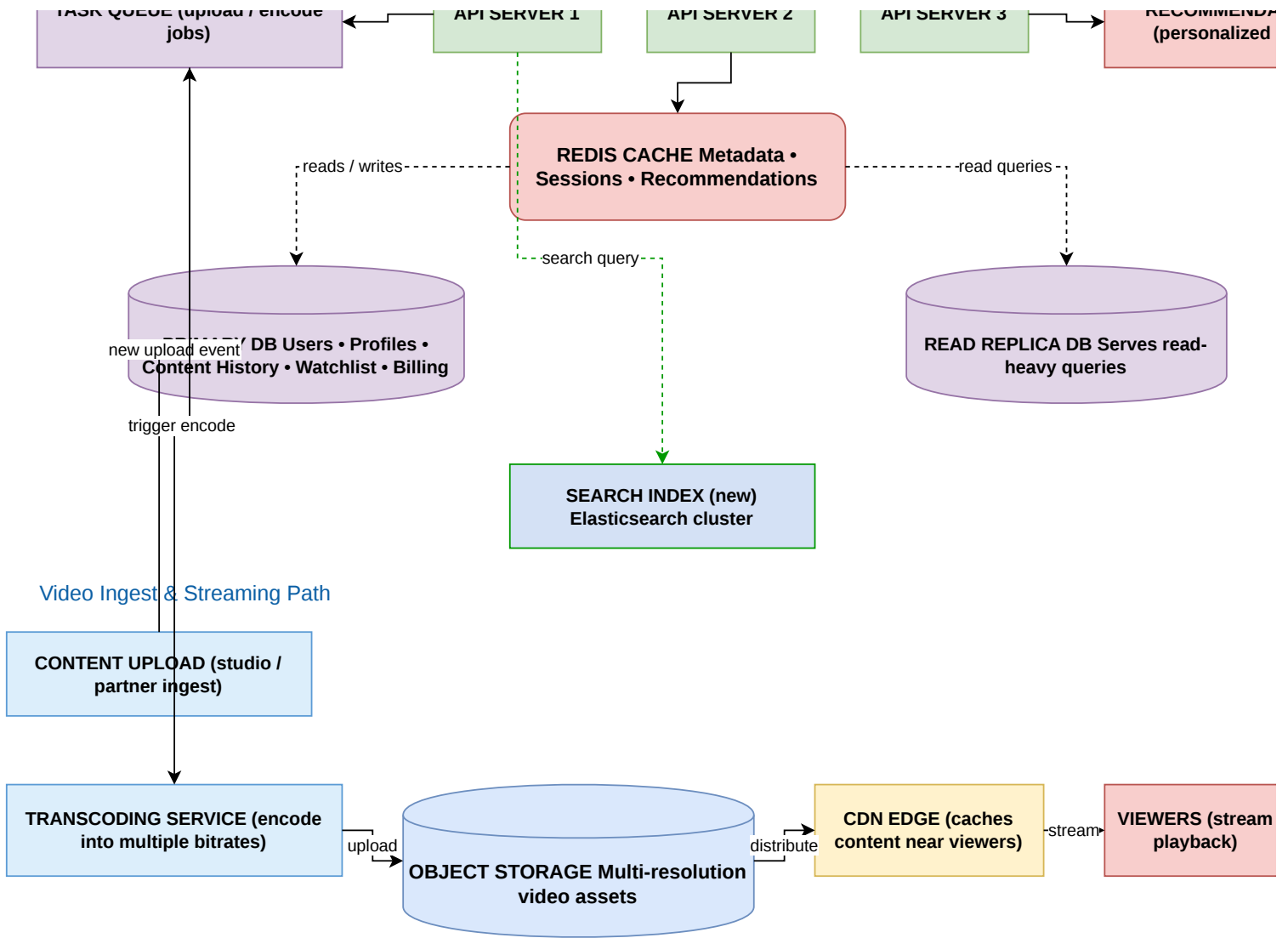
country	is_kids_profile	last_watched_at	duration
• USER (1) -> (N) PROFILE — one account can host several viewer profiles • PROFILE (1) -> (N) WATCH_HISTORY — each profile accumulates its own history • CONTENT (1) -> (N) WATCH_HISTORY — a title can appear in many history entries • PROFILE (N) -> (N) CONTENT — via WATCHLIST (many-to-many) • USER (1) -> (1) BILLING — one account maps to one active billing record			
BILLING (new)			
billing_id (PK)			
user_id (FK)			
plan_tier			
next_charge_date			
payment_method			

5. API Design

Method	Endpoint	Purpose
POST	/api/auth/login	Authenticate a user and issue a session token
GET	/api/catalog/browse	List titles by category, genre, or trending list
GET	/api/catalog/search?q=	Search the catalog by keyword
GET	/api/catalog/{title_id}	Fetch full detail page for a title
POST	/api/profiles	Create a new viewer profile under an account
GET	/api/profiles/{id}/history	Fetch watch history for a profile
GET	/api/recommendations/{profile_id}	Fetch personalized title suggestions
POST	/api/watchlist	Add a title to a profile's watchlist
DELETE	/api/watchlist/{title_id}	Remove a title from the watchlist
PUT	/api/playback/progress	Update resume position for a title
GET	/api/stream/{title_id}	Fetch a signed streaming manifest / CDN URL
GET	/api/downloads (new)	Request an offline download package

6. Architecture Diagram





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